Data Versioning & Lineage

RecoMart Recommendation Pipeline | Task 8 Deliverable

1. Versioning Workflow

Data versioning is implemented to ensure reproducibility and traceability of pipeline outputs. Every file produced by the pipeline is hashed using SHA-256 and tracked in a central versions.json manifest. This allows us to identify exactly which data was used to train a given model.

1.1 Version Tracking System

Location: data/versioning/versions.json

For each dataset (CSV/Parquet) generated during preparation and feature engineering, the system captures:

- Unique cryptographic hash (SHA-256) of the file contents
- File size, row count, and column count
- Timestamp of generation and logical version number

1.2 Data Lineage

Location: data/versioning/lineage.json

A Directed Acyclic Graph (DAG) representation is maintained to track dependencies between artifacts. For instance, the lineage graph records that product_features.csv is derived from prepared_products.csv, mapping the flow of data across pipeline stages.

2. Current Version Manifest Snapshot

The following artifacts are currently tracked in the versioning manifest (Sample):

Artifact	Row Count	SHA-256 Hash (Truncated)	Version
product_features.csv	1,018	87aad92d3f8510b56a9e71fef7d5928...	v1
product_features.parquet	1,018	e435d5817dbb1c2aa251ef93f16b5f5...	v1
user_features.csv	200	ea5902005e82e8e1be8d2e1beaab557...	v1
user_features.parquet	200	6c5fcbb9f1d4ae1e832441490559830...	v1